CONSTRAINTS ON THE EMERGENCE AND INTERPRETABILITY OF LINGUISTIC CONVENTIONS

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Abstract

Human communication takes place across many circumstances, and humans are adept at flexibly using language to achieve goals even in constrained conditions. The formation of linguistic conventions is often studied in iterated reference games, where over repeated reference to the same targets, a describer–matcher pair establishes partner-specific shorthand names for targets. In this dissertation, we expand the range of empirical evidence about when conventionalization occurs and what types of conventions are formed, all within the framework of iterated reference games. In Chapter 2, we ask which properties of the group's interaction structure facilitate successful communication, using a repeated reference game paradigm where we manipulated several key constraints on the group's interaction, including group size, the amount of feedback that matchers could give to directors, and the availability of peer interaction between matchers. In Chapter 3, we explore the developing pragmatic and referential communicative abilities of young children using a simplified, child-friendly paradigm. In Chapter 4, we consider the nature of conventions in iterated reference games, in particular to what extent the partner-specificity of these linguistic conventions causes them to be opaque to outsiders. Overall, we find that some signatures of convention formation can occur across a wider variety of circumstances than previously known, and that the descriptions used in iterated reference games are relatively comprehensible to outsiders.

Dedication

To those who supported me in this journey.

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The path my PhD and my PhD years took was shaped by a number of people to whom I owe thanks for their support.

First, I want to thank my parents for supporting me throughout my entire life. They cultivated intellectualism and knowledge-seeking in me from a young age. They also showed me that going to graduate school (at Stanford) was just a thing one does; the Stanford part is mostly coincidence.

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I'm extremely grateful to Mike for all of his support and wisdom. I heard a lot of praise about him when I was choosing a graduate program, and I think it's all true and then some. I've learned a lot, both from direct pedagogy and from observing what Mike does. And not just about research and academia, but also (and this is I think especially by example) how to be a good person, balance life and work, and deal with the ups and downs of life in academia. (Sorry about writing somewhat sappy things Mike, but I really do appreciate your being there for me throughout my PhD.)

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Introduction: Computational psycholingistic approaches applied to understanding convention formation in iterated reference games

2.1 Motivations and definitions

The use of language is a feature that distinguishes humans from other species. Language enables coordination between individuals, enforcing social norms via gossip, and communicating complex and abstract ideas, which in turn allows for the cultural accumulation of technologies beyond what any individual could invent. Paired with the technological innovation of writing, language allows for communicating across long distances of space and time.

Humans and other social species can communicate in a variety of modalities, but linguistic structures make language more expressive than other species' communication systems. Humans learn languages in early childhood and then are able to use the shared set of symbol-meaning mappings and combinatorial rules in order to communicate with other humans. Even young children realize that language is a very powerful system for transmitting ideas from mind to mind, as they demonstrate when they use it to make demands upon their caregivers. As a literate human, language is ubiquitous to (at least my) daily life, but the more one considers language, the more there is to explain and the more impressive our linguistic capabilities seem.

The word “language” can refer to a suite of related concepts all of which are targets of study in related domains. There’s the abstract idea of a specific language, for instance, English as a set of grammatical rules and word meanings. This is an abstract idealization as attempts to specify formal rules that cleanly generate the accepted utterances of a language run up against variability across communities. From a statistical perspective, one can aggregate across variability to study language as it is used by a certain community, not a static entity but something that evolves slowly over time. However, this is also an abstraction, as individuals each have separate grammars that may not fully align even within a community, so meanings must be treated as somewhat probabilistic or distributional. The idea of an individual’s grammar is an abstraction over the process of language learning from individual exposures to speech acts (and texts) and the psychological processes and neural architectures that produce and process language.

Separate from the scale at which we study language, there are other lenses that language is studied through. Language is preserved in artifacts of text and recordings that are studied separately have the human context and intentionality that produced them. Language can also be studied as a capacity for learning a system for expressing ideas and then expressing and understanding ideas in a system. From an information-theoretic lens, language can be viewed as a particular communication channel, subject to the same bounds as any other signal-transmission method. From the opposite perspective, languages often have substantial social and cultural relevance to their users. Generalizing across purposes, language is a tool that humans use to communicate with each other, and we can use both the combinatorial structure of language and pragmatic variations on the structure to accomplish a variety of communicative goals via the meaning conveyed and other features that are implied.

These different aspects of language reveal the complexities of studying language and lead to disagreements around what is studied in linguistics or the “language sciences". In the dissertation, I focus primarily on language as a tool for communication, but I will also consider other aspects of language including artifacts of language, language as a shared set of expectations, and psychological processing of language.

Another consideration when studying language use is what counts as language? Some scholarly traditions concentrate on conversational face-to-face language use for communicative purposes, and exclude derivatives like wordplay, phonetic codes, or language models as non-linguistic things that happen to use linguistic symbols. In this work, I do not assign a priori primacy to any particular parts or modalities of language. It may be that some types of language use are easier to process than others, or that some uses of language rely on domain general inferences external to the language faculties, but these are empirical questions whose results I will not assume.

While not as polysemous as the word `language`, `pragmatics` also has a number of partially overlapping meanings. When I use `pragmatics` I mean pragmatics expansively as something along the lines of meaning in context or context as a component to inferring meaning (Bohn et al., 2019).

This dissertation focuses on on how humans use language to communicate around the edges of expressivity – how we extend our lexica to accommodate communicative needs, and how we coordinate with each other in these atypical uses of language. These linguistic stretches can be ad hoc and temporary, but if they fill appropriate communicative niches, what are initially atypical usages can spread and lead to the adoption of new words, new meanings and other forms of language change.

As previewed above in discussion of the many angles from which language can be studied, there are trade-offs between studying language in more realistic situations and studying language in more tractable and controlled environments. In this work, I focus on language use *in context* because it is both tractable enough that experiments can usefully inform models and theories and analogous enough to real-world situation to have some explanatory power.

In the rest of the introduction, I first motivate the use of iterated reference games as a model system. I then discuss three main approaches from psycholinguistics that can be applied to convention formation in iterated reference games: efficiency, Bayesian pragmatics, and information theory. Then I review findings from the conversational and iterated reference game literatures and consider how these findings might be subsumed into larger psycholinguistic frameworks. Finally, I close with avenues for further inquiry and preview the few avenues that are addressed in the rest of the dissertation.

2.2 Iterated reference games as a model system

Iterated reference games are a commonly used paradigm for studying adaptive language use. A *reference game* consist of a set of people communicating with each other about a set of potential referents. One person, the *describer*, has a specific designated target (usually pre-identified as the target by the experimenter) that they seek to communicate to the *matcher* (or matchers). This act of reference requires coordination between describer and matcher on a symbol – meaning alignment. *Iterated* reference games involve repeated reference to the same targets within the same group of people, so that a certain conversational history builds up that could shape the construction and understanding of later symbols. Crucially, this task uses referents that are somehow difficulty to name in context, usually by a combination of low-codeability or high similarity to each other.

As described, this task is agnostic to communication modality, and iterated reference games are sometimes conducted using gestures or drawing (Bohn & Frank, 2019; Garrod et al., 2007; Hawkins, Sano, et al., 2023). Here, I focus on how this task interacts with the rich baseline of symbol – meaning mappings afforded by natural language.

People sometimes use words in creative ways to express meanings. For instance, even if a person doesn't know the name for an object or is having a tip-of-the-tongue event, they can still communicate about the object by using a more circuitous label. If this communicative need occurs repeatedly, an initially ad-hoc referring expression can calcify into a settled name for the object. Communicative acts serve a variety of purposes, but a simple, yet central one is reference: connecting a symbol with a meaning to pick out a target. This is a subgoal needed for more complex acts – it's hard to tell a story or debate someone without agreement on what different words or phrases refer to.

Combining these observations – that language can be adapted to fill communicative needs, that repeated need leads to changes in referring form, and that reference is a easy to elicit communicative act that is central to other communicative acts – suggests the utility of iterated reference games.

Iterated reference games are a commonly used task. The earliest studies we are aware of are from the 1960s (Krauss & Bricker, 1967; Krauss & Weinheimer, 1964, 1966), and iterated reference games seem to have become more popular starting in the 1980s (Clark & Wilkes-Gibbs, 1986). Iterated reference games have been used to study a wide variety of communicative and linguistic phenomena. The primary targets of study are coordination and the formation of conventions or conceptual pacts [(**dahan2025?**); Ghaleb et al. (2024); Eliav et al. (2023); Hawkins, Frank, et al. (2020); Metzing & Brennan (2003); Brennan & Clark (1996); among others]. Some studies focus particularly on iterated reference games as a way of studying collaboration in natural conversations (Clark & Wilkes-Gibbs, 1986; Dahan, 2024; Schober & Clark, 1989; Wilkes-Gibbs & Clark, 1992). Other studies focus on other aspects of the coordination, including the negotiation of referring expressions (Bangerter et al., 2020; Knutsen & Le Bigot, 2018) and the role of feedback (Fox Tree & Clark, 2013; Krauss & Glucksberg, 1977; Krauss & Weinheimer, 1966). Iterated reference games with multiple partners are used to study how speakers design their utterances for their specific partners (Horton & Gerrig, 2002, 2005; Turner & Knutsen, 2021; S. O. Yoon & Brown-Schmidt, 2014, 2019a, 2019b) and how speakers might generalize patterns of social conventions across interlocuters (Hawkins, Goodman, et al., 2020; Hawkins et al., 2021). Iterated reference games are used as the dialogue condition in studies on the differences between monologue and dialogue (Branigan et al., 2011; Fox Tree, 1999; Murfitt & McAllister, 2001; **tolins2017?**). Iterated reference games have also been used to study conversation and memory (McKinley et al., 2017; Vanlangendonck et al., 2018), model cultural conflicts between merging teams (Weber & Camerer, 2003), examine speech-gesture tradeoffs (de Ruiter et al., 2012) and study the effects of alcohol on social dynamics (Garrison et al., 2024).

In addition to studies on typical populations, iterated reference games are also used to study the abilities of special populations. Iterated reference games are used with children and teenagers (Branigan et al., 2016; Leung et al., 2024; Turner & Knutsen,

2021) and with older adults (Horton & Spieler, 2007; Hupet et al., 1993; Lysander & Horton, 2012) to study cognitive and linguistic abilities across the lifespan. Iterated reference games are used to study types of memory in people with amnesia (Duff et al., 2005; Duff, Gupta, et al., 2011; Duff, Warren, et al., 2011; S. O. Yoon et al., 2017) and to study various populations with neurological and psychological conditions Chantraine et al. (1998).

2.3 Frameworks

Just as one might hope that the results observed in iterated reference games generalize to more realistic situations, one also might hope that the results of iterated reference games could be explained by broad coverage theories of language use. I review three broad approaches that are common within computational psycholinguistics: efficiency, Bayesian pragmatics, and information theory.

2.3.1 Efficiency

One unifying framework gaining traction in psycholinguistics is efficiency, the idea that language and language use is under pressure to support efficient communication by maximizing the ratio of relevant information transmitted to effort. Efficiency is a high level framework that can only be tested against data when it is fleshed out with linking assumptions; however, comparisons between language use that is attested versus counterfactual languages can bound what types of assumptions are needed for this framework to be parsimonious.

Efficiency is thought to arise from trade-offs between communicative expressivity and some combination of learnability and ease of production (Kirby et al., 2015; Piantadosi et al., 2012). The state of language is thus in a dynamic equilibrium between the usage pressure to be able to communicate many ideas (pushing complexity of language) and some constraints pushing for simplicity of language, which could be either the requirement that languages be learnable or pressures for ease of production.

Evidence for efficiency comes from the argument that features of language are distributed much more closely to the Pareto frontier than would be expected by chance. A historically well-known example is that word frequencies follow a power-law distribution, which Zipf (1949) explains in terms of a "principle of least effort", where words that are more frequent are also shorter. It is worth noting that this type of power-law distribution is common across domains and generated by a variety of processes, so while well known, it is not the strongest evidence for efficiency as a driver of linguistic distributions (Piantadosi, 2014). Stronger evidence comes from the lexical partitioning of subdomains such as color, number, and kinship terms, where the distribution of systems falls on the Pareto frontier between complexity (number of terms) and informativity (how many bits each term provides) (Gibson et al., 2019; Kemp et al., 2018; Zaslavsky et al., 2018). Syntactic features of language such as harmonic word order and dependency length also appear to be optimized for increased expressivity with minimized processing effort (Gibson et al., 2019; J. Hawkins, 1995).

Efficiency arguments are based on the language artifacts of grammars and transcripts, but efficiency pressures act on language use as a process, not language as a static code (Gibson et al., 2019). Thus efficiency can be seen as imposing a joint constraint on the entire communicative process: minimizing the total time and effort involved in going from an idea in one person's head to a sufficiently close idea in another person's head. A corollary of this framing is that while shorter utterances are usually considered more efficient, utterance length in syllables or clock-time is not actually what is under pressure, they would not be efficient if they took longer to produce or parse than the more verbose alternatives.

As with other high level frameworks, the parsimony of the efficiency framework is entirely based on its auxiliary linking theories (Gershman, 2018). Thus, the question becomes what sort of auxiliary assumptions are needed to connect a given phenomena under the efficiency umbrella, and how reasonable are those assumptions.

As an illustration, I look at the issues of ambiguity and redundancy in referring expressions.

Redundancy and over-informative referring expressions

How do we reconcile the idea that language is efficient with violations of efficiency in language use ("look at the yellow banana") and the ubiquity of ambiguous utterances?

Terms such as "redundant" and "over-informative" are commonly used to describe situations where people produce modifiers that do not restrict the extension of a noun phrase, e.g. "blue cup" when only one cup is salient. In this example, we might think that the substring "cup" is a alternative utterance that could uniquely identify the target (Rubio-Fernandez et al., 2021). People do produce these non-restricting modifiers in reference tasks, especially for color, but this behavior seems to run counter to the idea of efficient language use.

Formalizing claims of redundancy or over-informativity requires a definition of what would be minimally informative, which in turn depends on a commitment to a fully specified semantic-pragmatic system, at least for the range of utterances to be considered. In the more general case, this quickly gets quite thorny: for instance, if specificity implicatures are within the option space, are those calculated before or after informativeness is measured (Bergen et al., 2016)? One could sidestep the theory issue by empirically measuring the information content of different utterances by how they shift the entropy of the distribution of inferred meanings (Degen et al., 2020), but this does not scale up well in part due to issues with compositionality.

The flip side of "redundancy" is ambiguity: many, many utterances are ambiguous. In general, strong contextual factors render the ambiguity a non-issue (Piantadosi et al., 2012), but this means we can't judge language outside of the physical and social context it is used in. Determining what is efficient language use in context requires not just analyzing phrases and their alternatives, but also how long utterances take to generate and comprehend, which may be highy contingent on contextual factors and conversational history.

The idea that utterances should have "just enough" information (Grice, 1989) has inspired a wealth of empirical research into what utterances people produce and what utterances people comprehend. By comparing these two halves of language use, we can determine how calibrated the utterances people produce are to what other people will understand. If calibration is reasonable, that would support an efficiency framework with minimal linking assumptions; if the calibration is not, we would need to invoke other assumptions (for instance, bottlenecks on production, goals such as politeness rather than just informativity) and then consider whether they are still parsimonious.

2.3.2 Bayesian pragmatics

Another broad coverage framework in cognitive science is modeling humans as Bayesian agents making resource rational choices. Within language, this tends to show up as a Bayesian pragmatic models, where (boundedly) rational Bayesian agents reason about each other to determine the meanings of utterances.

A prominent model within Bayesian pragmatics is the Rational Speech Acts (RSA) framework, an information-theoretic, computational framework for making quantitative predictions about pragmatic inferences in contexts (Frank & Goodman, 2012; Goodman & Frank, 2016). The basic idea of the RSA family of models is to picture two interlocuters recursively reasoning about how the other would produce or interpret utterances, grounding out in a listener (or speaker) who behaves in a pre-specified "literal" way.

Computational frameworks such as RSA provide a way to factor together different trade-offs and determine their relative weights in a model (Goodman & Frank, 2016). A softmax is taken over the scores of the options to produce a distribution of interpretations and utterances, with some parameters such as the degree of optimality fit based on data. This framework is usually run with one or two levels of recursion, where it tends to produce a reasonable fit to human experimental judgments, consistent with work finding that most people reason pragmatically at a low recursion depth (Franke & Degen, 2016).

RSA models have been used to predict some instances of ad hoc pragmatics as well as conventionalized pragmatic implicatures (Bergen et al., 2016; Degen et al., 2020; Goodman & Stuhlmüller, 2013). Models generally include a utility or informativity term that relates to how well an utterance resolves uncertainty in favor of the target referent. It is common to also include factors such as the prior likelihood of referring to each target (salience prior) and some cost on utterances where longer or more complex utterances are penalized (Goodman & Frank, 2016). Some models also go beyond informativity, incorporating options to infer the question-under-discussion (Kao, 2014; Qing et al., 2016) or balance informativity with politeness (E. J. Yoon et al., 2020). RSA-type models have also been used as a model of word learning in children (Bohn et al., 2022; Ohmer et al., 2022).

A full RSA model would incorporate all of these components and infer their weights. However, for tractability, usually only those features that are considered relevant to the domain of interest are included. Because of the flexible framework, it is possible to model many sources and levels of uncertainty, and then integrate out that uncertainty to make predictions, but also update on the sources of uncertainty in response to input.

The Challenge of Semantics

Perhaps the largest challenge to RSA models is the question of how to ground out the models in a "literal" listener or speaker. For the most part, RSA is tested in toy domains where the set of possible utterances is small and it is possible to enumerate a set of meanings (Bergen et al., 2016; Frank & Goodman, 2012; Goodman & Stuhlmüller, 2013). In some domains, a soft or continuous semantics is used to represent that some dimensions of meaning might be more strongly informative than others (Degen et al., 2020). This semantics supports the prediction of patterns of "redundant" color adjective use in referring expressions.

Soft semantics can run into conflict with compositionality. One can have soft semantics without explicit compositionality, if there is a way to determine match between expression and meaning. One could use a pre-trained neural network system that embeds utterances into semantic space and then taking distances to the target meaning, although this requires making assumptions about what "literal" meaning is. The system might have implicit compositionality, but there are no guarantees about the relationships between the embeddings of "red", "apple" and "red apple" in these systems. This may be more severe for expressions like "green apple on the red box" and "red apple on the green box" where syntactic relations are key to meaning which sentence-level embeddings may not fully preserve (Erk, 2022). Depending on the domain, embedding systems may or may not have satisfactory match to human semantic judgments.

Alternatively, one could have compositionality, but this requires a process for mapping the meanings of pieces into the meanings of the whole. The process could be deterministic rules, or an algorithm or neural model implementation, but one still needs a way to break down the units of meaning and build them up. In formal semantics, this is often door with a system of logical relations and lambda calculus which can assign that "red apple" is the intersection of "red" and "apple". However, this would then predict that "apple" is equally true for members of "red apple" and "blue apple", which from Degen et al. (2020) doesn't match people's judgments due to typicality effects.

One reason to want both compositionality and soft semantics is to be able to learn and update the prior on the basis of interactions. Given that an expression – meaning pairing was used successfully (or unsuccessfully), one may want to locally update the lexicon to reflect this additional information.

In less toy domains, there is not a satisfactory answer: some situations can be handled by empirically measuring likelihoods in an exhaustive ways, but this holistic approach is not compatible with incremental RSA (Cohn-Gordon et al., 2018) or larger sets of utterances that require compositionality to be defined. Some work has used grounded neural language models for alternative generation and likelihood measures, which has scaling potential in domains that have the right grounded datasets (Cohn-Gordon et al., 2018; Monroe & Potts, 2015; White et al., 2020). In order to extend RSA towards more realistic and open-ended scenarios, an important question to grapple with is what form of meaning (even at a computational level) will appropriately support pragmatic reasoning.

2.3.3 Information Theory

Information theory is a formalism that is used with both Bayesian pragmatics and efficiency as a link to connect abstract ideas to concrete measures and predictions. Information theoretic concepts such as entropy, surprisal, and rate distortion theory have been key to quantitative theories in computational work on language.

The ideas of information theory provides a link from functionalist accounts of language usage to the artifacts of language systems, by focusing on what the constraints are that would render the language systems we observe an optimal solution to the functional question of communication (Futrell & Hahn, 2022). One advantage of information theory is that it provides ways to calculate measures of complexity in a mostly theory neutral way (Futrell & Hahn, 2022). Information theory based theoretical proofs and toy models have been applied to domains such as language evolution (Plotkin & Nowak, 2000).

Measures derived from information theory such as surprisal, conditional entropy, and mututal information have been very productive in language science both for unifying aspects of language typology (ex. dependency locality) (Futrell & Levy, 2017) and for explaining patterns in language processing (ex. surprisal theory and entropy reduction) (Hale, 2001, 2016; Levy, 2008). The idea of rate distortion and a limited channel compressing a meaning through a message has been key to testing efficiency claims related to syntax, word order, and grammatical markers (Futrell & Levy, 2017; Mollica et al., 2021).

There is also a connection between information theory and pragmatics, as RSA formulations are closely related to solutions to information-theoretic joint optimization between utility and effort (Zaslavsky et al., 2020).

One difficulty with applying information theoretic measures is that it requires a probability distribution to work with; thus, similarly to other frameworks, there are questions of what distributional language models or datasets to use to capture the baseline language.

2.4 Phenomena

The communication and conversation literature, especially with regard to referring expressions, has provided useful descriptive characterizations of how people communicate in naturalistic and experimental settings. For iterated reference games in particular, there is an accumulation of fairly consistent and reliable findings of reduction, convention, and partner-specificity. While there are consistently observable trends in the conversation literature broadly and in iterated reference games specifically, there has not been strong identification of necessary and sufficient conditions or dose-response relationships between circumstances (or experimental settings) and the size of the effects. Thus, there are not quantitative accounts for the phenomena. Instead, the effects tend to be characterized with verbal theories that posit causes.

It is difficult to directly test these theories as they often fail to make explicit predictions about what would happen under conditions other than those tested or what experiments would adjudicate between theories. Many of these theories present certain results as “special” but our question as we go will be to what extent these phenomena could be explained by the broader coverage theories of efficiency and rational communication.

As I go through different theories and phenomena, I will consider the available evidence, problematize the verbal theories, and attempt to see how the above approaches from computational psycholinguistics might be used to more formally model or subsume the empirical results.

2.4.1 Mentalizing and non-mentalizing approaches

One historically important distinction that recurs with conversation, and interactions between agents more generally, is whether and how agents track other agents' internal states of knowledge and how this factors into their interaction. While an intuitively appealing distinction, it's not clear how to test this.

One could build, for instance, model-based and model-free algorithms and then compare which better captures an observed phenomenon, but this seems to harken back to associationist versus rule-based accounts in linguistics, which don't seem as relevant now. These aren't opposites so much as points on a spectrum and appearances may also be deceiving – what looks rule based could be instantiated in a non-symbolic, associationist way. On the other hand, a more sophisticated approach could look more associationist under performance load.

While is it not clear how to test these theories, it is still worth discussing the frameworks for historical context.

Mentalizing tradition and "common ground"

The "mentalizing" tradition treats humans as representing other humans as agents with internal states that include knowledge and goals. Within this broad school, there is variation in how these representations are implemented, how information gets added or modified, what exactly is tracked, and when representations (versus heuristics) are used.

Within this tradition, many use the term "common ground" to refer to knowledge that two agents share. In some cases, it is used in a pre-theoretic way to mean roughly "things you think another person will understand and won't be surprised if you reference" (Garrison et al., 2024; Leung et al., 2024). For instance, Hanna et al.

(2003) defines common ground as the “mutual knowledge, beliefs, and assumptions” held by the interlocuters. This meaning is roughly comparably to “givenness” in other domains (Fay et al., 2010).

However, “common ground” is also sometimes used in a theoretically loaded way, originating from the privileged versus mutual versus common knowledge framework (Clark & Wilkes-Gibbs, 1986). Under this usage, “common ground” is defined via infinite recursion in knowing that the other person knows that the first person knows that ...; this is the usage that comes up in formal semantics where many things must be introduced to common ground via accommodation, which could itself be seen as a kludge to explain how people get away with making statements whose presuppositions aren’t known to be shared (Horton & Keysar, 1996; Pickering & Garrod, 2004). In practice, humans don’t tend to do more than a couple layers of recursion in their pragmatic reasoning (Franke & Degen, 2016). Thus, it is generally not important to distinguish knowledge types at deeper recursion levels than mutual knowledge that both people know to be mutual.

Even if we accept that common ground has grown away from its formal semantic roots to just be a convenient shorthand for “mutual knowledge”, there is still a question of how agents know (or decide) what is mutually known. Many approaches have tried to characterize when something is mutually known (Brown-Schmidt, 2012; Clark & Wilkes-Gibbs, 1986; Horton & Keysar, 1996). This has predominately taken a deterministic approach with rules such as “if it’s in shared visual presence, it’s mutually known”. Visual presence may be a good heuristic, but it seems like whether something is mutually known may be better represented in a more graded way.

These terms seem to be easily subsumed within a Bayesian inference and Bayesian pragmatics framework. While the “mentalizing” approach places special emphasis on “common ground”, it’s unclear whether this form of modeling another’s knowledge should be treated differently from cases where one models another’s knowledge that one does not personally have. For instance, in deciding who to learn from, someone might determine that another agent knows what’s behind a door and then use this information to for instance, interpret a quantifier like “some” (Goodman & Stuhlmüller, 2013).

Another implication from the Bayesian approach is that we do not need to deal with deterministic rules and can instead reason under uncertainty. This may be useful for accounting for hedging behavior where there seems to be uncertainty over what an interlocuter will understand. Joint updating can also help for times when something was initially thought to be likely in the shared prior, but then new information decreases this likelihood. The framework of flexible inference and probabilistic knowledge states is a powerful tool.

Non-mentalizing approach: Interactive alignment theory

In contrast to mentalizing approaches, "interactive alignment theory" attempts to explain how people can successfully collaborate on reference tasks without reasoning about each others' mental states (Gandolfi et al., 2022; Pickering & Garrod, 2004). The motivation for non-mentalizing accounts is the apparent difficulty of mentalizing, coming from accounts such as naive egocentrism (which will be discussed more later). Given that humans reason socially about each other readily and from a young age (Rakoczy, 2022), it's not clear how well the motivation for a non-mentalizing approach holds up.

Regardless, at the time, there was desire for an approach to explain communication and coordination behavior without positing explicit reasoning about other minds, because it was that was too computationally expensive and slow. Under this framework, Pickering & Garrod (2004) tried to account for observed alignment between interlocuters by extending ideas from structural priming.

Structural priming occurs when processing a grammatical structure (for instance, a double object construction) increases the likelihood of expressing an idea using that structure rather than an alternative (producing a double object rather than a prepositional object construction) (Tooley & Traxler, 2010). In production, this occurs even when there is not lexical overlap (ex. different verbs), but is increased if there is lexical overlap. A similar effect occurs in comprehension, but it seems to only happen when there is the same lexical head item (Tooley & Traxler, 2010). Many potential mechanisms behind structural priming have been suggested, but there is not a consensus. While typically observed in controlled settings, the tendency to reuse constructions to express ideas is also observed in dialogue (Pickering & Ferreira, 2008).

Some other forms of alignment in dialogue can be observed in natural corpora. Interlocuters on Twitter exhibit style alignment although the direction of accommodation is moderated by social factors which suggests it may not be fully automatic (Doyle et al., 2016). Analyzes of Twitter and Switchboard corpora reveal lexical repetition, but not category alignment, between interlocutors for non-content word categories (Doyle & Frank, 2016).

Pickering & Garrod (2004) claims that the alignment occurs via “priming” and is “resource-free and automatic”, without providing a further explanation of what this means or how this works in terms of memory and processing. Unfortunately, as expressed, interactive alignment theory doesn’t provide an algorithmic or mechanistic level theory. There are a few issues with this theory. For one, it is unclear that structural priming would predict this cross-level trickle-up or trickle-down effect that is key to interactive alignment; the claim that people tend to say “the ball that’s red” after their partner says “the block that’s blue” is quite different from a claim that this also aligns their “situation models”. Another issue is that it’s really hard to test for intent (or lack thereof); for instance, repetition could be due to some sort of “priming”, or explicit reasoning, or production-side ease-of-use constraints (MacDonald, 2013).

Perhaps the largest problem with interactive alignment theory is that it does not account for one of the key features of iterated reference games: that people converge to conventionalized referring expressions that evolve and shorten after both people understand each others expressions. Thus, one requires an explanation for change, not just repetition or mimicking. To my knowledge, interactive alignment does not have an explanation for this, in part perhaps because their studies tend to be of a different sort. The “common ground” tradition generally using asymmetric director/matcher designs (dating back to at least Krauss & Weinheimer (1966)) and the “interactive alignment” traditions using symmetric designs where people both have partial information about a maze that they navigate together. Thus it is possible the two approaches are built around trying to explain differing sets of experimental results.

Figuring out exactly what alignment happens in conversation may well be worthwhile, but it does not seem that useful for the phenomena in iterated reference games in particular. That said, it is reasonable to be concerned that full inference is not necessarily cognitively feasible. One way to explore this would be with approximations of full Bayesian inference in the resource-rationality tradition or with information theoretic limits on memory capacity that don't allow for full partner representation.

2.4.2 Partner specificity

One key phenomenon from iterated reference games is that describers change their behavior when the set of matchers changes, usually shifting back to longer descriptions when a new matcher is introduced. This change in behavior is described as "partner-specificity", with the idea being that the conventional names developed with one partner as specific to that partnership (Brennan & Clark, 1996; Hawkins et al., 2021; Metzing & Brennan, 2003). The idea of partner-specificity is also referenced with regard to how different pairs diverge to different names for the targets (Hawkins, Frank, et al., 2020). Partner specificity is part of the mentalizing tradition and assumes that partners (and their background and knowledge states) are being represented in the minds of speakers. The form of the representation is not explicitly stated, so these models could be compatible with heuristic or distributed representations, but include explicit thinking about the audience and are incompatible with the non-mentalizing "priming" account.

Empirical evidence from experiments where one director talks with multiple partners suggests that people do "partial pooling" over their partners (Hawkins, Goodman, et al., 2020; S. O. Yoon & Brown-Schmidt, 2019a). That is, a speaker A will show some variation in their expressions when talking to partner B versus partner C, but there will be some generalization between partners as well, so that A talking with B is more like A talking with C than D talking to E. When coupled with a tendency for descriptions to shorten within a pair, this leads to a jagged pattern of reference length: when switching to a new partner, speakers use longer utterances, but not as long as their initial utterance with their first partner (S. O. Yoon & Brown-Schmidt, 2019a).

This pattern could be efficient if the context from the initial utterances is requisite to understand the later parts. The jagged pattern in particular is suggestive of a mix of speaker production effort and shaping the matcher's lexicon. If we consider the elaborated and shortened forms of the concepts (ex “arms up, like a goalie diving for a ball” versus “goalie”) to be synonyms (in the same way chimp and chimpanzee are synonyms), then the switch to short form after establishment and the temporary switch back with a new listener could be explained under a uniform information density model, provided that the longer form occurs in a context when it is more surprising and informative (Mahowald et al., 2013).

Partial pooling and partner specificity can be instantiated in a hierarchical model of Bayesian pragmatics, such as those introduced in CHAI or lexical uncertainty models (Bergen et al., 2016; Hawkins, Franke, et al., 2023; Potts et al., 2015; Schuster & Degen, 2020). Testing out these model frameworks with natural language data is not yet tractable due to the need to figure out computational and semantic issues, but the models seem to provide coverage at least in theory.

How can we think of partner specificity from an information-theroretic lens? One approach might be to think about what the mutual information is between the conversation history, the current utterance, and the visual context of targets. Within a partnership, the conversation history can serve as a sort of encoder that allows later conversational elements to use a more efficient coding to map between language and referent. Under partner specificity, one would say that conditioned on conversation history, the target and convention predict each other, but that conditioned on no history or a different conversation history they would not predict each other. This abstracts over having a model of how conversation history unfolds, but it’s at least an operationalization that could be testable using machine learning models.

2.4.3 Audience design & effort sharing

One set of theories in communication that are often tested and explored with iterated reference games have to do with how two (or more) people split the communication load.

One relevant concept is “audience design”, the idea that speakers seem to be sensitive to the knowledge state of their listener and say things that are easy for the listeners to comprehend. This seems to occur sometimes in everyday life where we speak differently to a 5 year old than to an adult, or explain things differently to a colleague than to a lay person. However, despite the introspection that we design utterances for the audience in some cases, it is difficult to identify if it is happening intentionally in referring games.

Confusingly, the phrase "audience design" sometimes implies intention on the part of the speaker (Horton & Gerrig, 2005; **horton2002a?**), and sometimes is used when utterances are constructed based on what's easy for the speaker, and listener ease is a side effect (Horton & Keysar, 1996; MacDonald, 2013; Rogers et al., 2013). For instance, audience design could both occur in times when the speaker gives an elaborated description to a naive listener (inferred to be intentional if contrastive with their description to a non-naive listener), but speakers may also tend to start descriptions with given material which is both more accessible to the speaker and convenient for the listener.

Audience design is sometimes used very broadly (V. S. Ferreira, 2019) when production choices make it easy for listeners to comprehend, even if this is not customized to the listener. In a very broad sense, efficiency pressures for things like uniform information density could be construed as audience design. Beyond iterated reference games, speakers do behave differently depending on the listener; speakers tell a story differently when retelling it to the same listener versus when telling the story again to a new listener (Galati & Brennan, 2010).

Intention versus side-effect are difficult to distinguish between because speakers and listeners often share recent context, find the same things salient, and linguistically what is easier to produce is often easier to process. Thus, disentangling speaker and listener ease may require careful experimental designs where ease of production and ease of comprehension are separated (F. Ferreira, 2004).

Questions around audience design are related to larger issues of how interlocuters split the communicative burden with one another. Depending on the task and the communication modality, there may be many options for how to balance the communicative load (Clark & Wilkes-Gibbs, 1986; Fay et al., 2010; Fox Tree & Clark, 2013). For instance, a listener could describe what options they see or otherwise prompt the speaker. We might expect the load splitting to vary based on the capacities of the interlocuters (ex. a speaker might craft their utterances more when talking to a child versus an adult) and the capacities of the channels (ex. speakers may use different approaches if listeners can interrupt).

While much of the work on iterated reference games and communication more generally has focused on interactions that are dyadic at any given time, much conversation in the real world takes place in larger groups. Multi-way conversations complicate the theories of audience design and partner specificity by introducing a larger audience of more partners. Two main questions are whether speakers "aim low" or "aim high" in balancing the needs of the listeners and whether speakers track individual listeners or an aggregate (S. O. Yoon & Brown-Schmidt, 2017, 2019a).

Empirical results indicate that speakers are sensitive to the knowledge states of listeners in a gradient way. Describers seem to track the knowledge states of different matchers independently when they alternate between addressing matchers (S. O. Yoon & Brown-Schmidt, 2017). When addressing a group of matchers with disparate knowledge levels all at once, speakers generally give descriptions that are fairly similar to what they give only naive listeners (S. O. Yoon & Brown-Schmidt, 2017). However, there is also some gradient, where descriptions vary based on the proportion of naive listeners in groups of up to 4 addressees (S. O. Yoon & Brown-Schmidt, 2019a). With pairs of listeners who differ both in their background knowledge and in what context they are seeing the target in, speakers produce descriptions that are sensitive to the interaction between background knowledge and distinctiveness of the target in context (S. O. Yoon & Brown-Schmidt, 2019b).

Speakers also use strategies in group contexts that don't occur as often in dyadic contexts, such as referring to a target with both the name that one person will understand and a elaborated description that will help another person get on the same page (S. O. Yoon & Brown-Schmidt, 2019a). The ability to track partner's knowledge states presumably would degrade as groups got bigger, but this paradigm has not been used for groups large enough for this to happen.

One angle on larger groups is network structures, where (non-linguistic) convention formation has been studied on networks of up to 50 people (Guilbeault et al., 2021). In this communication game with targets varying over continuous space, large groups tend to end up with fairly consistent category boundaries across independent networks, while small networks can support more idiosyncratic category boundaries. This work is suggestive that there may be group-size and group/network-structure dependencies on what sort of conventions arise. There has been some work on group and network designs in traditional iterated reference games, but only on relatively small groups.

One question related to partner specificity and audience design is whether the observed behavior is optimal. People are generally fairly pragmatic in their communication, so we might think that using different descriptions based on the composition of listeners is adaptive and calibrated. Speakers behave as if the descriptions that are appropriate to give vary based on the listeners knowledge state and their shared conversation history.

One source of evidence about the potential optimality is whether other people (naïve matchers) can understand the descriptions, and whether this varies by circumstance. This has not been addressed from the efficiency and optimality angle, but the role of dialogue and conversational history has been studied. Naïve matchers tend to do better the more their observation history resembles that of the original conversation—when hearing descriptions in order instead of in reverse order (Murfitt & McAllister, 2001), when listening to the entire game instead of starting in the third round (Schober & Clark, 1989), and when seeing yoked trials from a single game rather than trials sampled across 10 games (Hawkins, Sano, et al., 2023). Descriptions produced in dialogue are easier to understand than those produced in monologue (Murfitt & McAllister, 2001). The gradient suggests that some of the development in descriptions may be adaptive, but there has not been work grappling with whether the far above chance performance of naïve matchers poses problems for audience design theories.

It certainly suggests that from a purely audience design listener-focused perspective, speakers are not efficient, in that to the extent that new people can understand later utterances without context, a counterfactual more rapid adoption of short forms could achieve a certain level of accuracy just as well as the observed pattern. This is an area that could benefit from empirical work around what the optimal pattern could be.

There are also some alternative auxiliary hypotheses that could save the efficiency idea before we are forced to conclude that humans are miscalibrated. One possibility is that the communicative bottleneck is firmly on the production side – speakers are doing the best they can to rapidly produce descriptions, and producing more succinct descriptions earlier would be too costly in terms of effort or time. Another possibility is that social norms and task dynamics dissuade the use of short form descriptions even if they are likely to be understood: perhaps the initial descriptions are more polite, or perhaps people are (overly) cautious in their efficiency/accuracy trade-offs.

Bayesian pragmatic models could be extended to cover situations discussed here. Rather than just communicate pragmatically with one person, balancing multiple people's needs would require balancing different goals. Again the difficulty of specifying base interpretations and compositionality and reasoning may pose some difficulties, but a toy model of the situation could work. Similarly, a lexical uncertainty Bayesian pragmatics model would be an approach to capturing the new listener behavior in eavesdropper scenarios. In terms of effort splitting, that could also at least in theory be represented by changing RSA parameters such as depth of reasoning, starting lexical semantics, or cost functions.

2.4.4 Convention formation

One of the key phenomena of interest with iterated reference games is that over repetitions with the same partner, dyads tend to form shared "conventions" (also called "conversational pacts") about how to refer to the initially ambiguous targets. These conventions tend to be partner- and context-specific: changes in the speaker, audience members, or changes in the context can all license the use of a new description (Ibarra & Tanenhaus, 2016; Metzing & Brennan, 2003; S. O. Yoon & Brown-Schmidt, 2014). Absent these changes, pairs tend to stick near the semantic space of successful descriptions.

What exactly does convention formation refer to? There is ambiguity about what level of specificity convention formation and conceptual pacts refer to. It could be on the lexical level, such as calling a figure "ballerina". It could be conceptualizing the figure as a ballet dancer with a tutu (manifesting in descriptions with semantic association, but not lexical overlap, such as "ballerina" and "dancing in a tutu"). It could also be a general paradigm for how to describe figures, such as in terms as humans in different postures. (**horton2002a?**) distinguish between "lexical entrainment" when the same words are reused, and "conceptual similarity" when there is broader similarity that does not repeat the same words. These levels often co-occur, but in order to computationally quantify the phenomenon of convention formation, we have to make choices about how to operationalize the similarity: is it word overlap, lemma overlap, distance in semantic space, or something else.

In addition to convention indexing the stickiness of some properties of the description, there is also a pattern of change from typically a description of a set of features to a more holistic label. Typically, in convention formation, holistic or analogic descriptions are the ones that stick (Clark & Wilkes-Gibbs, 1986), but this isn't absolute. Groups can successfully coordinate on reference using many different types of names, including ones that pick up on low-level or meta-level features. The range of successful options makes explaining convention formation harder, and suggests suggests a strong path dependency for how reduced utterances evolve, potentially influenced by factors such as relationships and humor value. How to quantify this transition or model why it occurs are open problems.

The semantic meaning of a description is not a priori related to its length, but these two features empirically tend to correlate in iterated reference games (Hawkins, Frank, et al., 2020). Thus "reduction" or the shortening of utterances is sometimes used as a shorthand and measurement proxy for the semantic changes (Clark & Wilkes-Gibbs, 1986; Garrison et al., 2024; Hawkins, Franke, et al., 2023), and Bovet et al. (2023) considers it a measure of common ground. Convention and reduction are sometimes conflated with partner-specificity, as these phenomena often co-occur with different pairs forming different conventions and changes in group composition leading to (temporarily) longer descriptions (Clark & Wilkes-Gibbs, 1986; Wilkes-Gibbs & Clark, 1992). Better tools for measuring semantic distance separate from length may help address the theoretical issues of how entangled these concepts are. For instance, when a describer gives a more elaborated description based around the same conceptualization for a new addressee, where does that fall in terms of conventionalization?

It remains an empirical question whether the shortening of utterances, convention formation, and partner-specificity of descriptions are inseparable or merely occur together in the experimental paradigms considered in the literature.

From an information-theoretic perspective, we can think of convention formation as convergence towards a modified encoding that is more efficient at representing the specific set of target meanings than the baseline language. These are not meanings that are not usually commonly needed and thus don't have initially high codeability, but over local usage, a (temporary) warp of the meanings can be achieved that is locally more optimized to express the target meanings.

CHAI models

Scaling up RSA to handle reference games requires solving at least two problems. One is specifying a semantics system that can handle the abstract, metaphoric, and parts-based descriptions that are used – this could be seen as the problem of accounting for initial reference.

Secondly, one must explain how to go from one successful utterance to a different (often shorter) utterance. One RSA-style model that attempts to explain why multi-part descriptions are produced initially, but shorter descriptions are produced later is CHAI, a framework to bridge different levels of convention formation (Hawkins, Franke, et al., 2023).

CHAI incorporates parameterized hierarchical uncertainty over lexica that allow for variability in how words will be interpreted that can be integrated over. Listeners' lexica are not fully known which accounts for background differences that may be shared by a sub-population and differences from communication history that may be individual-specific. The results of inference update these priors, and propagate the new information to both the individual and group lexical representations. CHAI can qualitatively account for various convention-formation phenomena in toy systems (Hawkins, Franke, et al., 2023).

CHAI's explanation for reduction is consistent with the ambiguity arguments of Piantadosi et al. (2012): initially the context is not sufficiently constraining to resolve the ambiguity of a short utterance, so multiple descriptive pieces are needed to triangulate the meaning, but as conversational history accrues, the context is sufficient to disambiguate the shorter description.

In toy models of interlocuters playing a reference game with soft semantics, initial utterances use multiple properties to collectively increase the degree of certainty in the target. This successful reference then shapes the prior over word meanings, until the degree of certainty afforded by only one word is sufficient to pick out a referent.

The same approach CHAI uses to generalize from individuals to groups and populations could also be used to generalize across stimuli, if one could get the baseline semantics worked out. Then, a successful reference (like “man with arms up”) would not only update the meaning of that phrase or components, but could also lend weight to the over-hypothesis about using human and body part based descriptions, and thus update the speaker’s beliefs about the listener’s semantics for the words “head” and “leg”.

2.4.5 Developmental work

One angle to understanding linguistic phenomena is to look at their developmental trajectory to understand when the phenomena arises and how it changes across development.

There have been a limited number of direct studies of iterated references games with children, although there is a much wider body of work on young children’s communicative and pragmatic abilities.

One influential early study suggested that 4-5-year-old preschoolers struggle with child-child referential communication (Glucksberg et al., 1966). In their paradigm, one child was given a set of 6 blocks in a specific order. Their task was to describe the image on each block so their partner could pick out their corresponding block. As children described and selected blocks, they stacked them on pegs. While 4-5-year-old children succeeded on practice trials with familiar shapes and visual access to each other’s blocks, children failed on critical trials where the blocks had abstract drawings and there was no visual access. Even after multiple rounds with the same images, children were not able to correctly order the blocks. Glucksberg et al. (1966) attributed children’s communicative failures to their production of ego-centric descriptions that did not account for the other child’s perspective. Similar experiments with older children indicated a gradual improvement through adolescence both for initial accuracy and for the increase in accuracy across repetitions. Still, even the 9th grade sample was noticeably worse than the adult college student sample (Glucksberg & Krauss, 1967; Krauss & Glucksberg, 1969).

Since Glucksberg et al. (1966), few studies have revisited the question of whether children can successfully communicate in an iterated reference game. In one study, 8-10-year-olds exhibited adult-like patterns of increasing accuracy, increasing speed, and shorter descriptions across repetitions, but children's accuracy was still far below adult performance and highly variable between dyads (Branigan et al., 2016). 4-6-year-olds successfully used conventionalized gestures with their partners in an iterated reference game where children could only use gestures to communicate (Bohn & Frank, 2019). 4-8-year-old children succeeded at playing an iterated reference game with a parent using a simple, child-friendly tablet-based task (Leung et al., 2024). In this task, even 4-year-olds had an initial accuracy above 80%, which rose to above 90% in later repetitions (Leung et al., 2024).

One issue that arises is developmental work is assessing whether the paradigms are measuring the abilities of interest or whether task demands or cognitive limitations are masking nascent abilities. The overall trend in developmental work has been discovering evidence of abilities emerging earlier than previously thought, due to better methods and designs that require fewer additional cognitive resources. Thus, we might be skeptical of whether the negative results of Glucksberg et al. (1966) are an accurate reflection of children's communicative abilities. The perspective taking abilities of 4 to 5 year olds do seem to be greater than previously thought (San Juan et al., 2025).

What could psycholinguistic frameworks add to our understanding of developmental pragmatics? If we had a computational model of adult performance, the question would be what aspects to change or restrict to approximate child behavior. For an RSA type model, we could consider a lower optimization parameter, or reducing the amount of computation (sampling fewer options). Children might also do a similar process, but with a different size and baseline semantics in the lexicon. Would their updates in a repeated model be smaller or bigger? Even if we can't fit the model, the modeling formalism suggests ways of thinking about the differences between children and adults in a dimensional way.

If we take efficiency to mean an optimal tradeoff between expressivity and effort, then children might have different weightings between the two, ending up on a different frontier or a different part of the frontier. For instance, cognitive capacity constraints, or things like working memory or executive control could mean that certain types of utterances are particularly effortful, and thus shift the optimal curve to include less expressive utterances. For child – adult dyads, the shift in speaker / listener capacity may be substantially different than for adult - adult dyads, rendering different approaches more efficient. This may be difficult to test since it would require figuring how where the bottlenecks are empirically, or having good enough models to compare fits to the data.

From in information theory lens, we could conceive of communication channels involving children as being extra noisy or having a more limited channel capacity or both.

2.4.6 Incremental approaches to processing and production

Another angle from which to view reference is at the level of fine grained time course. As a referring expression is incrementally processed, what is the listener doing? How early do effects related to speaker identity or pragmatics occur?

The typical approach to understanding the unfolding of comprehension in real time is via visual world eye tracking where looks are interpreted as meaning to potential targets. Alternative methods to get at incremental interpretation and processing difficulty can also be used.

One idea that visual world approaches to reference try to address is whether there is an initial stage of interpretation that differs from the final resolution. For instance, certain influences like the literal meaning of the utterance might be available sooner than pragmatic or more contextualized interpretations. This is sometimes characterized as "top-down" versus "bottom-up" but there may also be different sources such as linguistic context or social context or visual access.

The relative balance and time course of top-down and bottom-up influences has been a big question in psycholinguistics and neurolinguistics for a long time (Gwilliams & Davis, 2022; Horton & Gerrig, 2005; Horton & Keysar, 1996; Tanenhaus et al., 1995). One debate in particular, in the (non-iterated) reference game literature specifically concerned the role of egocentrism in the early stages of linguistic processing. One side of the debate was the Egocentric Anchoring and Adjustment theory, where Keysar et al. (2000) argued for an initially egocentric perspective on the basis that people often initially look at objects that are good matches to a description even if the object is not mutually visible to the speaker Keysar et al. (2003). In particular, this theory claims a two-stage model where there is an initial (completely) egocentric perspective that can later be overcome, in contrast to the (much less controversial) idea that there is a general egocentric bias where one of the sources of information is egocentric but it can be in parallel to theory of mind information (Rubio-Fernández, 2008). The ego-centric first claim rests on a couple dubious linking hypotheses: that if people consider an interlocuter's perspective, their prior should be that the interlocuter only refers to mutually known things; and separately, that looking at an object is a sign that it is considered a potential referent (eye-tracking data is widely interpreted this way, but there is evidence that this proxy is only approximate, Degen et al., 2021). Their studies also always had the egocentric-only lure be extremely salient, by making it by far the best match to the description.

The counterpoint to initial ego-centrism presented by Hanna et al. (2003) is a constraint-based theory where many factors can play into comprehension. Many factors may influence language production and comprehension, to varying extents, and on differing time courses. Their studies use egocentric lures that are equivalently good to the shared visibility target and find early perspective taking, and confirm this with other experiments in different paradigms (Hanna et al., 2003; Nadig & Sedivy, 2002).

The larger picture of language processing and neuroscience strongly suggests that context information (at least from linguistic context) is used to predict upcoming linguistic material, and thus that "top-down" information can affect even early stages of linguistic processing such as phoneme identification and lexical processing (Gwilliams et al., 2023; Gwilliams & Davis, 2022). Predictive coding is a theory with good coverage of neural evidence in many domains, including language, with language processing regions seemingly sensitive to both lexical and structural information in predicting upcoming words (Shain et al., 2020). However, we might still wonder if earlier behavioral readouts are more influenced by perceptual information and less pragmatic, or if cognitive load also sways the balance of influence.

2.4.7 Production constraints

Compared to the theoretic landscape around psycholinguistic processing, production is much less studied because it's a lot harder to study experimentally.

There is agreement that production requires conceptualization, formulation, and articulation, but that is about all that is agreed upon (Bürki, 2018). From interference studies, we can get some sense of when certain elements in sentences are planned (Momma et al., 2016). On a word level, the phonological realization of words that are more frequent or usually appear in more predictable contexts is reduced relative to words that usually convey more information (Bürki, 2018). This could be consistent with either a production facilitation account or an audience design account – speaker and listener language statistics are similar so it is difficult ot disentangle. For production beyond the scale of single words, that are just more questions, including what the planning looks like, how the time courses of different stages of production may overlap, and what size and types of units are stored in memory (Bürki, 2018). There's a lot we don't know about the mechanics of production even in situations with less context-sensitivity than iterated reference games.

As with behavioral measures, neuroscience measures are also used more for processing than for production, and articulatory processes are better studied than the higher level planning processes. fMRI studies indicate that language production tends to activate the same regions as language processing (unsurprisingly), but these regions are especially active during phrase structure building, suggesting that production may have higher costs than comprehension (Hu et al., 2022). However, fMRI does not have precise time course resolution, so it can't resolve questions of how information flows in the brain.

The increased load from production is consistent with accounts that suggest that utterance design (and language evolution) are substantially shaped by biases to ease production. MacDonald (2013) points out three production biases: easy first, plan reuse, and reduce interference that could explain some patterns of utterance production, including in reference games.

One approach to identifying speaker-side difficulties in iterated reference games is to look for disfluencies (S. O. Yoon & Brown-Schmidt, 2014). Intuitively, producing descriptions of abstract figures is difficult and there may be load based on task difficulty that leads to non-optimal productions (ex using "square head" as part of a description when it is not diagnostic) that are reminiscent of children's describe/identify difficulties.

There are approaches one could take to try to understand more what production-side pressures are, including time-course locked measures such as eye-tracking and disfluency analysis. Carefully controlled studies and comparisons to monologue conditions may also help. Varying time pressure or pre-articulation planning time could also be used to help sort out what's going on (Horton & Keysar, 1996). But as the larger questions around production generally show, this is hard to study and may require very careful and clever experiment designs.

What do the broader coverage frameworks have to say about production? Production-side constraints are one of the factors posited to drive linguistic efficiency, but there aren't fleshed out predictions about speakers would or wouldn't say. In general RSA operates on a level of abstraction that only considers entire utterances and so only considers production cost in terms of the utterance sampling procedure (if not all options are considered) and the costs assigned to each utterance. However, given that most iterated reference games allow the speaker to keep talking, one could (again, in theory) adapt incremental RSA (Cohn-Gordon et al., 2018) to explain potential order effects within what parts of descriptions are produced first.

One information-theoretic computational-level model of incremental language production comes from Futrell (2023). This model applies the Rate Distortion Theory of Control to production, suggesting a trade-off between incremental steps that are good for the communicative goal (ex. being informative) with what is easy to produce, that is what is more automatic and predictable. As with other information-theoretic models, it has good explanations in toy or qualitative domains for explaining production ordering preferences, the use of filled pauses, and good-enough word production.

2.5 Where does this leave us?

A core theoretical question is whether the observed patterns of reduction, convention, and partner-specificity are due to specific mechanisms that only operate in a limited scope of iterated-reference-game-like situations or whether these results are explainable by broader coverage theories of pragmatic language use.

The phenomena observed in iterated reference games do not appear under other circumstances where people use pragmatic reasoning (including regular reference games). One possibility is that there are mechanisms specific to certain circumstances, and these mechanisms lead to the observed phenomena. For instance, some theories may focus only on conversational coordination with common ground and a backchannel. The exact scopes are often ambiguous, but they are not trying to account for all language use or even all pragmatic referential language productions. In some cases, the implicit scope may be smaller than the range of circumstances where the phenomena is observed, as is the case for theories that focus on dyadic mechanisms.

Alternatively, it could be that the same mechanisms that account for pragmatics broadly can also account for the phenomena in iterated reference games. The particular phenomena of interest in iterated reference games arise from general mechanisms playing out in certain circumstances such as when targets have low codeability and reference is repeated to the same target with the same partner. These circumstances are incorporated into the model as a parameters (what the targets are, what the contextualized lexicon is, what the communication channel is) and certain combinations of the parameters lead to the emergence of things like partner-specific convention formation.

Most of the iterated reference game literature has been focused on descriptive results and developing small-scope theories. Here I've been examining what sorts of parameterizations and auxiliary hypotheses would be needed for broad-scope theories to account for the results of iterated reference games. I believe our default hypothesis should be that convention formation emerges from the same mechanisms that account for all the rest of language use. As we attempt to figure out what set of mechanisms account for language use, iterated reference games are an interesting model for highly contextual language production and comprehension.

Having surveyed the phenomena and small-scope theories of iterated reference games, I now return to the three frameworks and synthesize how they might apply and what I think the interesting open questions are.

2.5.1 Efficiency

Sometimes, judging whether something is efficient or not may be clear cut, if all reasonable sets of assumptions return the same result. It is perhaps easier to judge something to be inefficient if there is a shorter alternative that is both easier to produce and easier to comprehend. In the general case, however, efficiency is very hard to cache out in specific predictions. What's efficient for an utterance in isolation may not be efficient when considered over an entire life of language use.

The formation of conversational pacts is often characterized as efficient, but this claim has not been cached out in formal models (Clark & Wilkes-Gibbs, 1986; Hawkins, Frank, et al., 2020). "Efficiency" is an informal description of these pacts that captures that shorter expressions are *more* efficient, but a given shorted expression could still be mis-calibrated to a particular situation. It could be too short, leading to misunderstanding, or still longer than it needs to be for effective communication. Mis-calibration of this type could contradict claims about partner specificity of reduction, but would not have to. Communicators could intend to be optimally calibrated but run into production difficulties, masking their competence. Either way, current claims of efficiency in conversation do not connect directly with the formal claims of optimal efficiency in the "language design" literature, and it is unknown whether speakers and listeners are calibrated or not.

Even with a specified task goal (communicate the target to a partner), what we tell people and what their actual utility function is are not the same. It's still a conversation between people so communication norms may also play a role, and that's difficult to control while retaining naturalism.

There are some open questions that are testable, mostly focused on figuring out what sorts of linking hypotheses would be required for the behavior to be (mostly) efficient.

One is figuring out how understandable utterances are from different points in the conversation, to basically test the idea that speakers are reducing "as fast as possible" for listeners; that is, both that extra words in an utterance are adding utility (adding information, adding certainty), and that the context of earlier utterances (and their repetition) is necessary for understanding later utterances (i.e. reducing faster would negatively impact accuracy). These are addressable questions, both with models and information theory applied to utterances and with testing new matchers on manipulated samples or orders of utterances.

The other issue is trying to get a handle on production-side demands. If speakers have the hard task, then perhaps what they say is mostly driven by ease of planning. This is harder to test, but one could do monologue studies and dialog studies with eye-tracking and time course measures to try to get at it in more fine grained detail.

Generally efficiency is usually applied to language the culturally evolved system based out of pressures that apply at each step. But it doesn't actually require that the individual steps be fully at the frontier, merely that there be enough pressure towards expressivity and ease overall that the system shifts in this direction. So, even if it turns out that people really don't seem efficient at this task, it doesn't hurt the efficiency framework that much because this task may be substantially out of distribution and thus, people's behavior on it is not shaping language use on a large scale.

2.5.2 Bayesian pragmatics

Bayesian pragmatics is a very promising large-scale framework for explaining a lot of linguistic behavior in context and incorporating history and non-linguistic context into a model of utterance production and comprehension. The main challenge is figuring out how to scale up to handle the natural language that comes up in reference games.

There are also some modeling considerations for how to extend models to deal with different types of data – how to deal with multiple listeners simultaneously, how to abstract across items to learn conceptual paradigms (i.e. abstracting from items to an approach of describing images in terms of what the person is doing). One question that comes up with Bayesian pragmatic models is how many free parameters they have and whether one is actually getting a parsimonious model with broad coverage, or just a set of related models very tuned to individual datasets. This is a broad issue in modeling and the closer we can get to a generative or cognitive model (even if fairly abstracted) the better.

So one way of making progress and seeing how broadly Bayesian models can be applied would be extending the existing models for a wider variety of designs. Group communication, either in the form of one speaker - multiple listeners (S. O. Yoon & Brown-Schmidt, 2014, 2019a, 2019b) or network structures (Hawkins, Goodman, et al., 2020), poses a challenge as there is not yet a non-dyadic extension of RSA that can do partial pooling across listeners and jointly optimize for the responses of multiple interlocuters. It should be possible to design extensions to CHAI to accommodate different group structures.

However, it would still lead to questions of whether the small-scale models can predict the patterns see in natural language, full scale experiments. And that requires solving problems related to the open vocabulary. How do you set baseline (literal) meanings for any utterance someone might produce, and how do you allow those to update in a sensible way? With toy models, one can do composition semantics and update based on the compositionality. While we could ground semantics in some neural model (ex. in some sort of VLM) that doesn't give composition units that can be individually updated. Thus, I think some of the challenge of getting a robust test of Bayesian pragmatics models requires a lot of NLP and ML engineering.

CHAI, while not a full explanation for the full patterns of real-world data, is a big step forward in at least providing a framework that actually explains why reduction would be optimal. It remains to be seen whether RSA-style models can predict the slope of reduction and the content of words that stay versus drop, but I believe the attempt would push forward our understanding of pragmatics and communication. Even without a cached out model, this approaches can inform how we think about experiments and results.

2.5.3 Information theory

Unlike efficiency and Bayesian pragmatics which are frameworks, information theory is more of a tool. It's robust to see that in iterated reference games, the utterances tend to get shorter, but what's actually going on with the language? In order to quantify that, it's useful to have some sort of metrics to use. One could theorize that words that are more informative are more likely to stick around and become the conventions. To test this, one would need a measure of "information" to apply, which may now be possible to test using computational models of language and stimuli.

Information theory also provides measures for trying to understand how context supports understanding. This could be to formalize ideas of common ground by considering the joint information between the conversation history (as the common ground) and the current utterance. Again, to measure this at scale we would need models to approximate the information, but there are now models that could be used.

The idea of task relevant information also makes this an interesting place to apply linguistic measures of information theory. For normal psycholinguistic texts, information theoretic predictors like surprisal or entropy reduction have substantial predictive value on processing difficulty as indexed by measures of reading time (or neural signals) (Smith & Levy, 2013; Wilcox et al., 2021). While surprisal is a measure of information, we might think that it's not the same as how many useful bits of information are conveyed. (Gibberish is quite high surprisal, but low information.)

One way that information theory might usefully frame what happens in iterated reference games is that the language is adapting to the specific task of optimally coding the target options. An optimal code would only need 3.5 bits to express each of the 12 options (a common number of options in iterated reference games). Many more bits than this of natural language are used in early rounds – suggesting that initially standard natural language is not a good fit for this task. Some of this language is conveying task-relevant information, just in a wordy way, while other parts may be not-task relevant at all (ex: describing features that do not vary across targets). Over time, people probably both drop some of the filler and identify the task-relevant components, but they also develop a more efficient code for those features. This formulation makes some testable predictions about the entropy of distributions over utterances at different points during the game.

2.5.4 Summary

From this survey of approaches as well as phenomena and theories, I think there are a few takeaways. I've highlighted some places where further work is needed to answer the question of how well broad-coverage psycholinguistic approaches explain iterated reference game results. Some of this is completely outside the reference domain – we could do with a better understanding of (general) language planning and production, and we could benefit from natural language processing models that do compositionality that could be plugged into Bayesian frameworks to interface with real data. There are also some computational and modeling directions, highlighted above.

One other direction is better characterizing the **explanandum**: what exactly are the phenomena we are hoping to characterize? While there have been many studies of iterated reference games across some variety of conditions, some theoretically relevant aspects of the space are under-explored. Some of the reference game specific theories focus on narrow parts of iterated reference game space, while all the broad coverage approaches expect a mostly continuous relationship between circumstantial knobs and the outcome measures. From a modeling perspective, having more data and a wider range of data also reduces some concerns that we are over-fitting to a very narrow circumstance.

Thus, I believe one necessity to substantial progress in the iterated reference game domain is a better empirical and descriptive understanding of the phenomena of interest. How do circumstantial knobs (group size, communication modality, stimuli, etc.) push around the extend to which we observe reduction, convention formation, and partner specificity? We don't have to finely map the entire landscape, but some sense of the geography of experimental space is necessary (Almaatouq et al., 2022).

More variable experimental circumstances could also address the question of how closely related the different phenomena of interest are. At least where it's been studied, reduction, semantic alignment within a group, and partner specificity have patterned together (to the point where some metrics are used as proxies for each other). How closely are they actually linked? Does reduction only occur via convention formation? A greater range of experimental situations could tell how well these measures separate versus correlate in different situations, which will in turn clarify what theory should explain.

2.6 Outline of the dissertation

In this dissertation, we expand the empirical foundation of iterated reference games in three directions, with the aim of addressing some theoretical assumptions and questions and providing some data for computational modeling in the future. We also apply some computational models to survey coarse-grained patterns of changes in semantic similarities.

In Chapter 2, we address the generalization patterns of the key patterns of results in iterated reference games, by examining the trajectories of convention formation in games of varying small group sizes and varying communication channel capacities. This expands the empirical foundation to mimic more of the different circumstances in which communication occurs in the real world, and allows for a test of how entangled the different reduction phenomena really are.

In Chapter 3, we turn to the developmental timeline of the communication and coordination skills embedded in iterated reference tasks. Here we re-visit historical claims that children are too egocentric to produce successful ad-hoc referential expressions, which is fairly incompatible with more recent work showing nascent pragmatic language use and communication skills in preschoolers. This different context again provides a test of how entangled different observed reduction phenomena really are and provides insights into how language use fits into broader cognitive and linguistic development.

Finally, in Chapter 4, we measure the level of opacity in the referential expressions produced in iterated reference games (using the corpus of reference games from Chapter 2). Taking inspiration from lexical uncertainty models, we use both naive human matchers and computational proxies to determine how far from "baseline semantics" referential expressions are at different stages of convention formation. These patterns are a test of at least the strongest versions of efficiency and audience design claims and give us a better understanding of what sort of partner-specificity is at play.

In the last chapter, I conclude with a discussion of where this leaves us and what directions seem most promising from here.

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